PRAYCG Master Comprehensive Analysis Suite v1.4.0

Deep Module Guide for TSC 2026 Review Conversations - v1.2 Formula Addendum

Detailed methods handout: what each module does, how it computes its endpoints, how to read the outputs, and what the suite can and cannot claim. Version 1.2 adds explicit custom composite formulas for artifact_score, API_A_v1, TSP peak/taper detection, and CandidateLocal_KHT event-lock logic.

Central boundary

This suite is an analysis and visualization system for PR-AYC-G pilot data. It can summarize EEG, physiology, timing, markers, and exploratory state-transition candidates. It does not prove consciousness, Opto-Structural Memory, hidden-Y biology, microtubular memory, biophotonic mechanisms, clinical efficacy, or a human EEG mechanism.

One-sentence operational summary

The suite takes a PR-AYC-G XDF/event-log/media-QC bundle, converts it into time-windowed EEG/autonomic/respiration/self-report features, compares Control/Target/Contextual Override, tests state-locked handoff candidates, estimates exploratory K_HT coupling, and writes auditable tables, figures, and synchronized review-video overlays.

Contents

1. Core experiment logic and analysis philosophy

PR-AYC-G branch logic

The three-arm design is simple: Control is a phase-scrambled version of the cue-embedded stimulus; Target is the intact narrative watched naturally; Contextual Override is the same intact narrative watched under analytic task demand. The suite asks whether Target differs from Control and Override after timing, artifacts, sensory load, task stance, respiration, and self-report are considered.

- Control asks: is the response merely light, sound, motion, cuts, cue timing, audio envelope, or sensory drive?
- Target asks: what happens when recognizable narrative meaning is accessible and the participant receives it naturally?
- Contextual Override asks: what happens when the same intact stimulus is present but the participant is forced into an analytic task stance?
- The strongest PR-AYC-G claim is not that a single signal moved. The stronger claim requires a pattern across EEG, physiology, timing, artifact controls, and self-report.

Rung 0O translation rule

The Opto-PING Rung 0O result changed the human analysis doctrine: prediction alone is not enough, and K alone is not enough. In PR-AYC-G, a candidate event should become interesting only when local coupling and theta handoff occur together, under timing and artifact constraints.

2. Inputs, outputs, timing, and evidence levels

Input	What it supplies	Common file/table
XDF recording	Raw LSL streams: EEG, HR/RR, respiration, ALS timing, StasisMarkers, status streams.	*.xdf
Event log	Local copy of protocol markers, phase starts/ends, ratings, cue markers, sum response.	*_events.json / *_events.csv
Cue schedule	Cue values, cue timecodes, expected override sum, media-prep provenance.	cue_schedule_*.json / .csv
Channel map	Electrode labels and ROI assignments. Required for scalp-level ROI grouping.	*_events_channel_map.csv
Media manifest/QC	Video hashes, Target/Override match, Control generation, ALS marker/prefix design, audio QC requirements.	media_prep_manifest_*.json

Annotation/anchor file	Predeclared media-structural anchors or secondary annotations for state-locked testing.	predeclared_anchor*.json / .csv
Feature table	Optional precomputed time-resolved features. If supplied, can speed visualization/analysis.	*feature_frame.csv

Evidence levels

Confirmatory requires predeclared media-structural or algorithmic anchors loaded before physiology analysis. Secondary uses annotation files or scene-level reports. Exploratory uses physiology-discovered peaks or top composite events in the same run. A physiology-discovered event can generate the next hypothesis, but it cannot confirm itself.

- ALS timing belongs to u(t), the external input/timing vector. It is not hidden Y, not K_HT, and not biological photonic data.
- TSP is a scalp-level posterior-temporal proxy. It is not STS/MTG source localization.
- K_HT is a human-translation coupling proxy. It is not K_OSM, Y_cell, Y_OSM, LTP, microtubular memory, or OSM proof.
- Self-report is independent evidence. It constrains and contextualizes physiology, but it does not validate the model by itself.

3. Feature extraction: from raw streams to analyzable windows

3.1 Windowing

- EEG is divided into sliding windows, default 2.0 seconds long with a 1.0 second step.
- Each window is assigned to a segment such as BASELINE_1, CONTROL_1, TARGET_1, CONTEXTUAL_OVERRIDE_1, WASHOUT_1, WASHOUT_2, or WASHOUT_3.
- Report-entry and sum-entry periods are censored using pre/post pads so typing, rating, and posture changes do not contaminate physiology windows.
- Washouts are trimmed at the edges by default, because the beginning and end of washout can contain transition artifacts.

3.2 EEG power and phase features

- For each EEG window, the suite detrends the samples and computes log bandpower using Welch power spectral density estimates.
- The suite computes power for theta, alpha, lower-gamma, and smaller 5-Hz lower-gamma bins.
- When requested, it computes phase/amplitude using bandpass filtering and Hilbert transforms, then derives phase-locking values (PLV) by ROI.
- PAC is exploratory only: theta phase to gamma amplitude modulation is summarized with a mean vector length style metric when the module is enabled.

Band name in suite	Frequency range	Primary use
theta_4_8	4-8 Hz	Integration/carryover and PNCC family endpoints.
alpha_8_12	8-12 Hz	Inhibition/settling context and feature layer.
g30_35	30-35 Hz	Lower-gamma sub-band for GammaScalpel.
g35_40	35-40 Hz	Lower-gamma sub-band for GammaScalpel.
g40_45	40-45 Hz	Lower-gamma sub-band for

		GammaScalpel.
lgamma_30_45	30-45 Hz	Composite lower-gamma work/meaning/task candidate band.
hf_proxy_45_55	45-55 Hz	High-frequency artifact proxy when available; not a meaning band.

3.3 ROI grouping

ROI label	PRAYCG16 example channels	Interpretation
frontoparietal_task	Fz, F3, F4, Pz, P3, P4	Task-lock / analytic-load candidate region.
meaning_candidate	Cz, C3, C4, T5, T6	Broad content-integration candidate set.
posterior_temporal	T5, T6	Posterior-temporal scalp proxy used by TSP.
visual_control	O1, O2	Visual/sensory control layer.
artifact_sentinel	Fp1, T3, T4	Blink/eye and jaw/temporal sentinel layer.
jaw_temporal_sentinel	T3, T4	Jaw/temporal muscle artifact guard.
primary_content_core	Fz, Cz, Pz, C3, C4, P3, P4, T5, T6	Broad non-sentinel content/core set.

4. Deep module reference

Module 1. Stream inventory and data integrity

Field	Detail
Purpose	Identify the XDF streams and determine whether the expected multimodal streams are present.
Primary inputs	XDF streams: EEG, StasisMarkers, OpenBCI status markers, ALS timing/AUX, PolarHRV or RR, respiration.
Main outputs	stream_inventory_and_timing_qc.csv; stream_inventory.csv; logs.

Algorithmic steps

1. Load XDF stream descriptors and sample arrays.

2. Record stream name, type, nominal sample rate, channel count, time span, and sample count.

3. Flag missing or unexpected streams; keep analysis modules from pretending absent data are available.

How to interpret it

- A stream being present is necessary but not sufficient; signal quality and timing still need separate checks.
- This module is the run receipt. It answers: what was actually recorded?

Boundaries / failure modes

- It cannot verify electrode contact, participant compliance, or signal validity by itself.
- An XDF with streams present can still have artifacts, timing gaps, or failed ALS pulse detection.

Module 2. Marker validation and phase segmentation

Field	Detail
Purpose	Turn StasisMarkers/event-log entries into analyzable protocol phases.
Primary inputs	Event log JSON/CSV and/or marker streams inside the XDF.
Main outputs	phase_intervals.csv; marker validation summaries; condition slicer intervals.

Algorithmic steps

1. Find phase start/end markers such as BASELINE_1_START/END, CONTROL_1_START/END, TARGET_1_START/END, and CONTEXTUAL_OVERRIDE_1_START/END.

2. Build segment intervals in LSL time.

3. Create derived windows such as final 20/30/60 seconds and washout splits when requested.

4. Prefer explicit event logs when present; otherwise use XDF marker streams.

How to interpret it

- This is the backbone of all condition comparisons.
- The visualizer uses these intervals to cut the selected branch from the full XDF when the user selects Control, Target, or Override.

Boundaries / failure modes

- Software markers record intended event timing; they are not physical display truth unless validated by ALS/photodiode.
- Missing end markers can force the suite to skip or downgrade a segment.

Module 3. Channel map validation

Field	Detail
Purpose	Associate EEG channels with electrodes and ROI labels.
Primary inputs	Channel map CSV or built-in PRAYCG16/MarkIV preset.
Main outputs	channel_map_used.csv; channel-map status in report.

Algorithmic steps

1. Load a supplied channel-map CSV if available; otherwise use a built-in preset.

2. Assign 0-based indices, electrode labels, ROI labels, and confidence metadata.

3. Pad or truncate to the number of channels actually present in the EEG stream.

How to interpret it

- High-confidence channel mapping is required before scalp ROI interpretations are taken seriously.
- The same numeric channel can mean different things if the cap was wired differently; this module prevents silent mismapping.

Boundaries / failure modes

- Channel-map confidence is not source localization. T5/T6 are scalp electrodes, not direct STS/MTG measurements.
- If the map is uncertain, ROI language should be downgraded to channel-level exploratory language.

Module 4. Cue schedule validation and override task scoring

Field	Detail
Purpose	Verify cue schedule provenance and score the Contextual Override sum task.
Primary inputs	Cue schedule JSON/CSV, event markers, media manifest, override response markers.
Main outputs	cue_schedule_validation_summary.csv;

	cue_marker_events.csv; contextual_override_task_scoring.csv.

Algorithmic steps

1. Read cue count, cue design, expected sum, cue position, and target/override hashes.
2. Extract cue markers from Target and Override event streams.
3. Score the participant-reported override sum against the expected sum.
4. Flag the result as exact correct, incorrect, or not scored.

How to interpret it

- Override task scoring helps interpret whether the analytic stance was active.
- An imperfect sum does not automatically invalidate Override if self-report shows high analytic effort and task compliance.

Boundaries / failure modes

- Task scoring is not proof that meaning was suppressed.
- Cue visibility and participant effort are separate from physiology.

Module 5. Self-report summary

Field	Detail
Purpose	Parse ratings for meaning, empathy, absorption, story-active washout, emotional afterglow, analytic effort, task compliance, and related constructs.
Primary inputs	RATING_* markers and RATINGS_*_END JSON notes in the event log.
Main outputs	self_report_wide.csv; self_report_long.csv; report sections.

Algorithmic steps

1. Find rating markers and/or final rating JSON notes.
2. Convert branch-specific subjective ratings into long and wide tables.
3. Summarize Control, Target, Override, and washout self-report patterns.

How to interpret it

- Self-report tells us whether the intended phenomenological state plausibly occurred.
- High Target meaning with low Control meaning supports the intended PR-AYC-G psychological separation.
- High Override analytic effort supports task engagement even if physiology remains mixed.

Boundaries / failure modes

- Self-report does not prove the physiological model.
- Absorbed subjects are not reliable clocks; scene-level notes are stronger than retrospective second-level timestamps.

Module 6. Artifact/input exclusion windows

Field	Detail
Purpose	Remove or flag periods dominated by reporting, sum entry,

	stillness-reset prompts, and other non-stimulus behavior.
Primary inputs	Event markers for report input, sum input, ratings, and return-to-stillness periods.
Main outputs	artifact_exclusion_windows.csv.

Algorithmic steps

1. Find report-input and sum-input start/end markers.
2. Pad report windows by default pre/post margins.
3. Skip EEG windows that overlap those exclusion intervals.

How to interpret it

- This prevents typing, posture changes, button presses, and report entry from masquerading as physiology.
- The report-input censorship layer is a methodological humility layer.

Boundaries / failure modes

- It does not remove all artifacts. Eye, jaw, motion, sweat, electrode drift, and respiration still require their own guards.

Module 7. EEG channel QC and feature extraction

Field	Detail
Purpose	Compute per-channel signal QC and windowed power/PLV/PAC features.
Primary inputs	EEG stream, phase intervals, exclusion windows, channel map.
Main outputs	eeg_channel_qc.csv; eeg_window_power_features.csv; eeg_window_power_plv_pac_features.csv.

Algorithmic steps

1. Choose the EEG stream and estimate effective sampling rate.
2. Within each clean segment window, detrend data and compute peak-to-peak amplitude.
3. Compute log bandpower for theta, alpha, 30-45 Hz lower gamma, and 5-Hz gamma sub-bands by ROI.
4. Optionally compute phase-locking by band/ROI and theta-to-gamma PAC.

How to interpret it

- This is the main feature generator. Most later modules read its columns rather than raw EEG.
- Peak-to-peak and high-frequency sentinel metrics later become artifact penalties.

Boundaries / failure modes

- Consumer EEG gamma is artifact-sensitive. Gamma features must always be interpreted with jaw/temporal, blink, high-frequency, and p2p guards.
- Feature extraction does not source-localize cortex.

Module 8. Mapped EEG contrasts

Field	Detail
Purpose	Compare windowed power and PLV metrics across PR-AYC-G

	branches.
Primary inputs	EEG feature frame by segment.
Main outputs	artifact_matched_condition_contrasts.csv.

Algorithmic steps

1. For ROI/band metrics, compare Target vs Control, Target vs Override, W2 vs W1, W2 vs W3, and final-window contrasts.
2. Sort windows by artifact score and match equal sample counts before computing mean deltas.
3. Bootstrap confidence intervals and sign-permutation p-values are reported as exploratory support statistics.

How to interpret it

- This module gives the broad first answer: which EEG metrics moved by condition?
- Artifact matching is designed to prevent dirty windows from dominating one side of a contrast.

Boundaries / failure modes

- A positive contrast is not a mechanism claim.
- This module does not know whether a gamma increase is meaning, task, muscle, sensory load, or artifact without the later layers.

Module 9. Residualized lower-gamma

Field	Detail
Purpose	Estimate whether lower-gamma condition effects survive partialling out nuisance proxies.
Primary inputs	Lower-gamma EEG metrics; artifact/sentinel/visual/task-related columns when available.
Main outputs	eeg_low_gamma_residualized_contrasts.csv.

Algorithmic steps

1. Identify lower-gamma target columns in primary/core, meaning-candidate, frontoparietal, and all-channel ROIs.
2. Regress or residualize against available artifact and nuisance features when the table supports it.
3. Re-test Target/Control/Override contrasts on residualized values.

How to interpret it

- Residualization is a guard against the simplest explanation: the effect was just global artifact or sensory drive.
- If residualized Target > Override is absent, gamma interpretations should be downgraded.

Boundaries / failure modes

- Residualization cannot rescue bad data. It also cannot prove causality.
- Over-residualization can remove real signal if the nuisance proxy is biologically entangled with the effect.

Module 10. GammaScalpel

Field	Detail
Purpose	Decompose lower-gamma into task-like, meaning-candidate, visual, and artifact-sensitive components using 5-Hz bands.
Primary inputs	Windowed power and PLV for g30_35, g35_40, g40_45, and

	lgamma_30_45 across ROIs.
Main outputs	gammascalpel_window_scores.csv; gammascalpel_score_contrasts.csv.

Algorithmic steps

1. Z-score relevant power and PLV columns at the window level.
2. Build an artifact score from peak-to-peak and high-frequency sentinel features.
3. For each gamma band, compute TaskGamma from frontoparietal power/PLV minus artifact penalty.
4. For each gamma band, compute MeaningGamma from meaning-candidate power while subtracting meaning-candidate PLV rigidity, visual-control power, TaskGamma, and artifact penalty.
5. Compare GammaScalpel scores across Target vs Override, Target vs Control, and final-window contrasts.

How to interpret it

- GammaScalpel treats gamma as work/load, not as direct meaning.
- TaskGamma rising in Override is expected if the analytic task successfully locks attention.
- MeaningGamma is a candidate when Target exceeds Control and Override after visual/task/artifact penalties.

Boundaries / failure modes

- GammaScalpel is heuristic and exploratory; it is not a neural decoder.
- Jaw/temporal muscle can occupy the same frequency neighborhood. Artifact veto has priority.

5. GammaScalpel in detail: what happens to each 5-Hz band

- The suite does not treat gamma as one undifferentiated number. It separately evaluates 30-35 Hz, 35-40 Hz, 40-45 Hz, and the 30-45 Hz composite.
- This matters because a broad 30-45 Hz increase can hide different patterns: one band may track visual binding, another task effort, another artifact, and another candidate semantic work.
- The 45-55 Hz band is treated primarily as high-frequency artifact proxy support when available, not as the main meaning band.

Band	What the module asks	Interpretive caution
30-35 Hz	Is there a lower-gamma work change in the low edge of the band?	Can overlap with broad cognitive load and residual motion.
35-40 Hz	Does the central lower-gamma region carry the effect?	Still artifact-sensitive; never read as meaning alone.
40-45 Hz	Does the upper lower-gamma edge carry a sharper transition/load signature?	Closer to EMG contamination risk; check jaw/temporal sentinels.
30-45 Hz composite	Does the lower-gamma system as a whole differ by branch?	Useful summary, but less diagnostic than the 5-Hz decomposition.

TaskGamma_band = mean(z(power_frontoparietal_task), z(PLV_frontoparietal_task)) - 0.25*z(artifact_score)

MeaningGamma_band = z(power_meaning_candidate) - z(PLV_meaning_candidate) - 0.25*z(power_visual_control) - 0.50*TaskGamma_band - 0.50*z(artifact_score)

Why subtract PLV for MeaningGamma?

The working assumption is that analytic/task locking can create rigid synchrony without narrative reception. A meaning-candidate score should not merely reward rigid phase order. Therefore the heuristic penalizes excessive same-band PLV in the meaning candidate layer.

Module 11. Task-lock PLV

Field	Detail
Purpose	Measure whether analytic/task stance creates increased timing rigidity across frontoparietal or core channels.
Primary inputs	Windowed phase-locking values by band and ROI.
Main outputs	Included inside EEG feature tables and contrasts; summarized in GammaScalpel and endpoint tables.

Algorithmic steps

1. Compute Hilbert phase for each selected frequency band.
2. Within each ROI and window, compute median pairwise phase-locking value.
3. Compare Override against Target/Control where relevant.

How to interpret it

- A high task-lock PLV can support that Override was cognitively engaged.
- It can also explain why Override gamma differs from Target: the same stimulus is being processed under a more rigid analytic stance.

Boundaries / failure modes

- PLV is not meaning. Rigid timing can reflect task effort, visual entrainment, artifact, or shared stimulus timing.

Module 12. PAC exploratory

Field	Detail
Purpose	Compute theta-to-gamma phase-amplitude coupling when explicitly enabled.
Primary inputs	EEG theta phase and gamma amplitude by ROI.
Main outputs	PAC columns in eeg_window_power_plv_pac_features.csv.

Algorithmic steps

1. Bandpass theta and gamma signals.
2. Extract theta phase and gamma amplitude using the analytic signal.
3. Compute mean vector length of gamma amplitude weighted by theta phase.

How to interpret it

- PAC can indicate cross-frequency organization in a window.
- It is useful as an exploratory check when theta/gamma coupling is central to a hypothesis.

Boundaries / failure modes

- PAC is sensitive to filtering, waveform shape, noise, and sample length. It is off by default and should not be elevated over the core PR-AYC-G contrasts.

Module 13. Autonomic API_A

Field	Detail
Purpose	Summarize heart rate and HRV/RMSSD into a rough autonomic availability/regulation index.
Primary inputs	PolarHRV/RR or HR stream and segment intervals.
Main outputs	hrv_api_segment_summary.csv.

Algorithmic steps

1. Detect whether the cardiac stream appears to be RR intervals or HR values.
2. Compute mean/median HR, RMSSD, SDNN, and absolute HR slope by segment.
3. Z-score these variables across available segments.
4. Build API_A_v1 as lower HR, higher RMSSD/SDNN, and lower HR slope weighted into one state index.

How to interpret it

- API_A helps distinguish regulated availability from mere arousal or strain.
- Target physiology should not be judged from EEG alone; the body is part of the receiver.

Boundaries / failure modes

- API_A is not a clinical autonomic diagnosis.
- RR artifacts, respiration, posture, strap fit, and movement can change HRV metrics.

Module 14. Respiration summary

Field	Detail
Purpose	Quantify breathing behavior and common-drive risk.
Primary inputs	Respiration belt stream and segment intervals.
Main outputs	respiration_segment_summary.csv.

Algorithmic steps

1. Extract respiration signal within each segment.
2. Estimate dominant breath-rate features and basic respiratory variability.
3. Report segment-level breathing metrics for control/common-drive interpretation.

How to interpret it

- Respiration can drive HRV and can modulate EEG rhythms, so it must be modeled or at least reported.
- A PR-AYC-G effect that disappears after respiration control is not a meaning-specific EEG finding.

Boundaries / failure modes

- Respiration summary is not the same as full respiratory phase correction.
- Noisy belts or posture shifts can make breath estimates unreliable.

Module 15. Formal PNCC_theta

Field	Detail
Purpose	Measure post-video theta carryover, especially whether the Target washout differs from Control/Override washouts.
Primary inputs	Theta PLV/power features in washout segments.

Main outputs	pncc_theta_artifact_matched_surrogate_contrasts.csv; pncc summary tables.

Algorithmic steps

1. Prefer theta PLV metrics; fall back to theta power if PLV is unavailable.
2. Compare WASHOUT_2 against WASHOUT_1 and WASHOUT_3, using artifact-matched surrogate contrasts.
3. Report deltas, confidence intervals, and permutation-style values.

How to interpret it

- Block PNCC_theta asks whether the after-state persists after the Target video ends.
- It is strongest when Target washout exceeds Control and Override washouts under comparable artifact conditions.

Boundaries / failure modes

- A video-end washout is not always the correct integration boundary; meaningful intake can happen earlier than the final frame.
- PNCC is compatible with the state-locked handoff family, but it should not replace it.

Module 16. tau_coh_theta

Field	Detail
Purpose	Estimate theta coherence lifetime / decay properties where the stream supports it.
Primary inputs	Theta PLV/coherence-like time series.
Main outputs	tau_coh_theta_summary.csv.

Algorithmic steps

1. Select a theta coherence or theta PLV proxy.
2. Model decay/autocorrelation-like persistence when enough clean windows are available.
3. Report whether the estimate is evaluable and the estimated coherence lifetime if available.

How to interpret it

- tau_coh_theta is the older coherence-lifetime endpoint family and remains useful for persistence logic.
- It is especially relevant to group/hyperscanning extensions.

Boundaries / failure modes

- Single-subject, short, noisy, or nonstationary data can make tau estimates unstable.
- A tau estimate does not prove intersubject synchrony in a single-person run.

Module 17. Surrogate theta carryover

Field	Detail
Purpose	Test whether theta carryover contrasts survive randomization/surrogate checks.
Primary inputs	Theta features, segment labels, artifact scores, random seed.
Main outputs	Surrogate contrast tables included in PNCC outputs.

Algorithmic steps

1. Build artifact-matched Target/Control/Override comparisons.

2. Bootstrap or randomize labels/signs within matched windows.

3. Estimate whether observed deltas are unusual relative to surrogate distributions.

How to interpret it

- Surrogates prevent over-reading small deltas that could arise by chance from unequal window counts or artifact distributions.

Boundaries / failure modes

- Surrogate quality depends on exchangeability. If windows are not comparable, surrogate p-values become weak evidence.

Module 18. Post-meaning frequency cascade

Field	Detail
Purpose	Look for ordered frequency changes after candidate meaning intake: high-frequency work, theta integration, and alpha/settling patterns.
Primary inputs	Windowed power and PLV features across bands and segments.
Main outputs	post_meaning_cascade_key_metrics.csv; post_meaning_cascade_order_index.csv.

Algorithmic steps

1. Extract key metrics around target, washout, final windows, or anchors depending on available features.

2. Compare whether the expected order appears: gamma/work before theta/integration and later settling patterns.

3. Report whether the classic or modified delayed-reveal pattern is supported.

How to interpret it

- This module asks about sequence, not just amplitude.
- A narrative response may not be a simple “gamma up, theta up” block effect; timing order matters.

Boundaries / failure modes

- Cascade logic is exploratory and can be sensitive to anchor placement and baseline choice.

6. TemporalSemanticProxy and TSP-to-theta handoff

Module 19. TemporalSemanticProxy (TSP)

Field	Detail
Purpose	Construct a posterior-temporal scalp proxy for narrative/semantic processing load, while penalizing visual, task, jaw, and global artifact contributions.
Primary inputs	EEG window features for posterior temporal, visual control, jaw sentinel, frontoparietal task, and artifact score.
Main outputs	temporal_semantic_proxy_window_scores.csv; TSP columns

	in feature frame.

Algorithmic steps

1. For each gamma band, start with z-scored posterior-temporal lower-gamma power.
2. Subtract visual-control lower-gamma when available.
3. Subtract jaw/temporal sentinel lower-gamma more strongly because jaw/temporal muscle can mimic scalp gamma.
4. Subtract frontoparietal task lower-gamma to reduce analytic-task contamination.
5. Subtract global artifact score.
6. Write TemporalSemanticProxy_g30_35, TemporalSemanticProxy_g35_40, TemporalSemanticProxy_g40_45, and TemporalSemanticProxy_lgamma_30_45 when available.

How to interpret it

- TSP is best read as "posterior-temporal scalp gamma after nuisance penalties."
- A TSP peak is an anchor candidate for semantic/narrative processing, not a source-localized STS/MTG measurement.
- The 30-45 Hz TSP column is the main TSP trace used by downstream TSP-to-theta handoff and CandidateLocal_KHT modules.

Boundaries / failure modes

- TSP cannot distinguish all semantic, social, visual, motor, and artifact sources by itself.
- If jaw sentinel or artifact_score is high, the TSP event is artifact-cautioned.

TSP_band = z(posterior_temporal_power_band) - 0.35*z(visual_control_power_band) - 0.50*z(jaw_temporal_sentinel_power_band) - 0.35*z(frontoparietal_task_power_band) - 0.25*z(artifact_score)

Algorithmic location of a TSP event

The suite does not ask the participant to timestamp the event. It smooths the TSP or gamma trace with a centered rolling median, finds the peak in whole-target, first-half, and second-half scopes, and then locates a taper point where the trace falls halfway back toward its local baseline. That taper is treated as a candidate boundary where theta carryover may begin.

half_taper_threshold = baseline + 0.5*(peak - baseline)

taper_time = first post-peak time where smoothed_trace <= half_taper_threshold; if not found, fallback = min(last_time, peak_time + 20 s)

Module 20. Peak and taper detection

Field	Detail
Purpose	Locate candidate intake peaks and post-peak taper boundaries that can be used as state-locked anchors.
Primary inputs	Target EEG feature frame with gamma, TSP, TaskGamma, MeaningGamma columns.
Main outputs	postpeak_peak_and_taper_detection.csv.

Algorithmic steps

1. Add relative time within each segment.

2. For Target, evaluate traces such as primary gamma, meaning-candidate gamma, posterior-temporal gamma, TSP, task gamma, MeaningGamma, and TaskGamma.

3. Evaluate whole-target, first-half, and second-half scopes.

4. Smooth each trace with a rolling median.

5. Find the maximum, estimate a local baseline, compute the half-taper threshold, and find the first taper crossing.

How to interpret it

- The peak indicates candidate intake/work; the taper indicates the possible handoff boundary.
- This is the central state-locked move: integration does not have to wait for the video to end.

Boundaries / failure modes

- If no taper is found, the fallback anchor is lower confidence.
- Because peak/taper detection is physiology-driven, it is exploratory unless the detection rule was frozen before analysis.

Module 21. PostPeakPNCC_theta

Field	Detail
Purpose	Measure theta carryover after a detected gamma/TSP peak has tapered.
Primary inputs	Peak/taper table and theta feature columns.
Main outputs	postpeak_pncc_theta_contrasts.csv; focused_postpeak_pncc_theta_results.csv.

Algorithmic steps

1. For each taper anchor, define a pre-window from taper-30 seconds to taper-5 seconds.

2. Compute theta mean in post windows 0-10, 10-30, and 30-60 seconds after the taper.

3. Compute Target pre-to-post theta delta.

4. Compute matching Control and Override deltas at the same relative times.

5. Report Target-minus-Control and Target-minus-Override deltas, plus artifact change.

How to interpret it

- This module asks whether theta rises after the candidate intake/work signal has released.
- It is the state-locked version of PNCC_theta.

Boundaries / failure modes

- A strong post-peak theta increase can still be non-specific if Control or Override also show it.
- A result is stronger when theta handoff follows Target more than Control and Override and artifact does not rise.

Module 22. AnnotationLockedThetaCarryover

Field	Detail
Purpose	Test theta carryover after predeclared or supplied narrative/media-event anchors.
Primary inputs	Annotation CSV/anchor file and theta features.
Main outputs	annotation_locked_theta_carryover_contrasts.csv.

Algorithmic steps

1. Accept eligible event types: semantic_reveal, climax, post_reveal_reflection, sustained_meaning_load, meaning_taper, threat_startle.

2. Use event end time as the anchor.

3. Compute pre-to-post theta changes in 0-10, 10-30, and 30-60 second windows.

4. When the anchor belongs to Target, compute same-relative-time Control and Override comparisons.

How to interpret it

- This is the confirmatory-friendly anchor family when events are media-structural and frozen before physiology analysis.
- It respects chronometric humility: freeze the scene, not the subject’s unreliable internal stopwatch.

Boundaries / failure modes

- Post hoc annotation files are secondary/exploratory unless the timing was frozen before analysis.
- The module does not infer meaning from the annotation alone.

Module 23. Gamma-to-theta and TSP-to-theta handoff

Field	Detail
Purpose	Estimate whether gamma/TSP predictors lead later theta outcomes at physiologically plausible lags.
Primary inputs	Gamma/TSP predictors, theta outcomes, artifact_score, Target windows.
Main outputs	gamma_to_theta_handoff_lag_correlations.csv; gamma_to_theta_handoff_best_lags.csv.

Algorithmic steps

1. Build predictor set: primary lower-gamma, meaning-candidate lower-gamma, posterior-temporal lower-gamma, TSP, and MeaningGamma when available.

2. Build outcome set: theta PLV/power in primary/core, meaning-candidate, posterior-temporal, and primary power when available.

3. For lags 2, 5, 8, 10, 12, 15, 20, 22, 25, 30, 40, 50, and 60 seconds, pair predictor windows with later theta windows.

4. Compute raw lagged correlation and artifact-partial correlation.

5. Report the strongest lag per predictor/outcome pair.

How to interpret it

- A positive handoff means the intake/work proxy tends to precede later theta integration rather than merely co-occurring with it.
- TSP-to-theta is the semantic-proxy version of the handoff test.

Boundaries / failure modes

- Lagged correlation is exploratory and cannot prove causality.
- The module is sensitive to autocorrelation and anchor-free search. Use predeclared anchors or random-anchor controls for stronger claims.

Module 24. Thresholded State-Update

Field	Detail
Purpose	Summarize candidate state-update sequences: high-frequency intake -> taper -> theta integration.
Primary inputs	Peak/taper table, theta metrics, Control/Target/Override features, artifact score.
Main outputs	thresholded_state_update_candidate_windows.csv; thresholded_state_update_summary.csv.

Algorithmic steps

1. Use peak/taper anchors as candidate state-transition boundaries.

2. For each anchor and theta metric, compare pre-window against 0-10, 10-30, and 30-60 second post windows.

3. Compute Target, Control, and Override deltas.

4. Grade an event as supported by sign when Target-minus-Override is positive and artifact is not rising; artifact-cautioned if artifact rises.

How to interpret it

- This module tries to detect thresholded timing: the state change occurs when the physiology transitions, not necessarily at video end.
- It is useful when subjective time is fuzzy or the meaningful moment happens before the washout.

Boundaries / failure modes

- It does not prove plasticity, LTP, or structural memory.
- It is exploratory unless the anchor rule and thresholds are frozen before analysis.

Module 25. State-Locked Handoff Alignment

Field	Detail
Purpose	Compare state-aligned anchors against video-end and random-anchor alternatives.
Primary inputs	Peak/taper anchors, annotation anchors, theta metrics, random seed.
Main outputs	state_locked_anchor_summary.csv; state_locked_epochs_long.csv; state_locked_model_comparison.csv; dtw_readiness_note.md.

Algorithmic steps

1. Build an anchor table from gamma/TSP tapers and annotation ends.

2. Extract state-aligned epochs around each anchor, typically before and after the candidate boundary.

3. Compare best state-locked theta delta against video-end theta delta and random-anchor distributions.

4. Write a DTW readiness note; constrained dynamic time warping is disabled by default until repeated same-stimulus or multi-subject data exist.

How to interpret it

- This module embodies the doctrine: integration boundary is not automatically the video boundary.
- It is useful for aligning data by the biological event rather than the MP4 clock alone.

Boundaries / failure modes

- State-locking can overfit if anchor discovery and testing occur in the same run.
- DTW can make bad matches look good; it should remain preconstrained and secondary.

7. HumanTranslation_KHT and CandidateLocal_KHT

Module 26. HumanTranslation_KHT

Field	Detail
Purpose	Estimate a rough phase-level reciprocal coupling between EEG gamma/work signal E_human(t) and a human translation proxy Y_HT(t).
Primary inputs	Time-resolved EEG features, constructed TSP/Y proxy, visual/task/artifact/theta controls, media covariate inventory.
Main outputs	human_translation_kht_feature_frame.csv; human_translation_kht_columns_used.csv; human_translation_kht_phasewise_summary.csv; human_translation_kht_cv_model_losses.csv; human_translation_kht_cv_win_summary.csv; human_translation_kht_interpretation.json.

Algorithmic steps

1. Standardize feature columns and normalize segment labels.

2. Choose E_human from available gamma/work features, preferring lower-gamma primary/core or posterior-temporal proxies.

3. Construct Y_HT from a direct TSP column when present, otherwise from posterior-temporal lower-gamma minus frontoparietal, visual, jaw, and artifact terms.

4. Compute V_proxy as the within-segment first difference of Y_HT.

5. For each segment, fit E_t from Y_{t-1}, E_{t-1}, and controls; the Y_{t-1} coefficient is eta_Y_to_E.

6. Fit Y_t from E_{t-1}, Y_{t-1}, and controls; the E_{t-1} coefficient is zeta_E_to_Y.

7. Compute K_HT as the positive square root of eta*zeta when the product is positive; retain signed square root separately.

8. Compare full reciprocal prediction against replay-only, sensory/task/artifact, generic-theta/artifact, and Y-proxy autoregressive alternatives with 5-fold blocked CV.

How to interpret it

- HumanTranslation_KHT asks whether a reciprocal human-level loop is visible at the phase/branch level.
- A pilot-positive pattern requires Target K_HT > Control and Target K_HT > Override, plus supportive model comparison and no dominant artifact/timing explanation.

Boundaries / failure modes

- K_HT is not K_OSM.
- Phase-level K_HT can miss brief local events, and local events can exist without phase-level endpoint confirmation.
- The module is exploratory in n=1 data and should be presented as a translation proxy only.

E_t ~ eta*(Y_HT_{t-1}) + beta_E*(E_{t-1}) + controls + error

Y_HT_t ~ zeta*(E_{t-1}) + beta_Y*(Y_HT_{t-1}) + controls + error

K_HT_positive_loop = sqrt(eta*zeta) if eta*zeta > 0; otherwise 0 or no positive reciprocal loop

Module 27. CandidateLocal_KHT

Field	Detail
Purpose	Estimate event-level local K_HT-style coupling around candidate anchors and require theta handoff to follow.
Primary inputs	Feature frame, predeclared anchors, annotation anchors, state-locked peaks/tapers, random-anchor reference windows.
Main outputs	candidate_local_kht_analysis.csv; candidate_local_kht_anchor_manifest.csv; candidate_local_kht_random_anchor_reference.csv; candidate_local_kht_interpretation.json.

Algorithmic steps

1. Build an anchor manifest from predeclared media-structural anchors, annotation-file anchors, state-locked/peak anchors, and top physiology-discovered composite candidates.

2. For each anchor, extract a local window around the event, default 10 seconds before to 10 seconds after.

3. Fit E_next from E_lag, Y_lag, and artifact; the Y_lag coefficient is local eta.

4. Fit Y_next from Y_lag, E_lag, and artifact; the E_lag coefficient is local zeta.

5. Compute K_local = sqrt(abs(eta*zeta)) and K_signed using the sign of eta*zeta.

6. Compute theta pre-window t-15 to t, immediate post t to t+10, and delayed post t+10 to t+30.

7. Sample random anchors within the same condition to build percentile references for K_local and theta handoff.

8. Declare final human-event lock only when the anchor is predeclared/media-structural and both local K and positive theta handoff exceed the p95 random-anchor reference.

How to interpret it

- This module is the most Rung-0O-like human module: K alone is not enough; theta handoff alone is not enough.
- It is well suited for future PRAYCG1.8B predeclared media-structural anchors.

- Physiology-discovered candidates can become future frozen anchors, not same-run confirmations.

Boundaries / failure modes

- Random-anchor percentiles are internal references, not population p-values.
- Event-level K_HT is sensitive to window length, feature choice, and local artifacts.
- CandidateLocal_KHT does not prove OSM or hidden-Y biology.

CandidateLocal_KHT decision rule

Final event lock = predeclared/media-structural anchor + local K significance + theta handoff significance. If the event was discovered from physiology in the same run, the claim level is exploratory even if the numbers look strong.

8. Additional modules and report-building layers

Module 28. Media covariate inventory

Field	Detail
Purpose	Record whether media fingerprints, manifests, cue schedules, and time-resolved media covariate tables are available.
Primary inputs	Media manifest JSON, stimulus fingerprint folder, cue schedule files.
Main outputs	human_translation_kht_media_covariate_inventory.csv.

Algorithmic steps

1. Check supplied media-manifest and stimulus-fingerprint folders.

2. Scan CSV/JSON candidates for time, seconds, frame, luminance, audio, RMS, optical flow, or cut-rate columns.

3. Record whether a time-resolved sensory covariate file appears available.

How to interpret it

- This module does not perform full sensory subtraction. It tells the analyst whether the ingredients exist.
- Time-resolved media covariates are required for stronger sensory-drive controls in later versions.

Boundaries / failure modes

- A file being present does not mean it was correctly aligned to the XDF.

Module 29. Master endpoint table

Field	Detail
Purpose	Collapse the many module outputs into a compact endpoint-status table.
Primary inputs	Core contrasts, residualized gamma, GammaScalpel, API_A, PNCC, tau, cascade, handoff/KHT summaries.
Main outputs	master_endpoint_status_table.csv; handoff_endpoint_status_table.csv.

Algorithmic steps

1. Read module-specific summaries and score whether each endpoint is supported, weak, mixed, not supported, or not evaluable.

2. Keep links to evidence tables so every claim can be traced backward.

3. Write a one-table summary for the report.

How to interpret it

- The endpoint table is useful for triage and conversation.
- It should not replace reading the detailed module tables.

Boundaries / failure modes

- A summary status can hide caveats such as dirty baseline, missing ALS pulse, weak self-report, or artifact concentration.

Module 30. Figures and report generation

Field	Detail
Purpose	Create readable summary plots and Markdown/HTML reports.
Primary inputs	Processed tables and feature frames.
Main outputs	figures/*.png; Master Comprehensive report Markdown/HTML/PDF when exported.

Algorithmic steps

1. Plot lower-gamma by segment when available.

2. Plot API_A by segment when available.

3. Plot respiration by segment when available.

4. Insert key tables into the report.

How to interpret it

- Figures are a review aid, not a statistical replacement.
- They are useful for spotting obvious condition patterns, timing failures, or questionable endpoints.

Boundaries / failure modes

- Plots can look persuasive even when the underlying endpoint fails model comparison or artifact veto.

Module 31. MasterSync Visualizer

Field	Detail
Purpose	Render a synchronized review MP4 with the stimulus branch on top and rolling physiological/EEG graphs below.
Primary inputs	XDF or feature CSV, selected stimulus MP4, event/anchor files, Master Comprehensive output folder.
Main outputs	review.mp4; *_features_used.csv; *_events_used.csv; *_render_report.json.

Algorithmic steps

1. User selects Control, Target, Override, or Full Session.

2. If XDF and branch markers are available, the visualizer slices the XDF to the matching branch interval.

3. If a feature CSV is supplied, it filters by condition and resets the selected branch to time zero.

4. It auto-adds CandidateLocal_KHT and state-locked overlays from the Master Comprehensive output folder when available.

5. It renders rolling panels for EEG theta/gamma, HR/HRV, API_A, respiration, ALS timing, and highlighted events.

How to interpret it

- The visualizer is excellent for communicating timing and inspection to third parties.
- It prevents accidentally pairing Target video with Override physiology when condition slicing succeeds.

Boundaries / failure modes

- It is a visualization/audit artifact, not an endpoint validator.
- A beautiful synchronized MP4 does not certify meaning, OSM, hidden-Y biology, or task compliance.

9. How to read the most important output tables

Table	What to look for	Main caution
eeg_window_power_plv_pac_features.csv	Time-windowed power/PLV/PAC columns by segment and ROI.	Feature table, not an interpretation.
artifact_matched_condition_contrasts.csv	Target-Control and Target-Override deltas with artifact matching.	Positive deltas are not mechanisms.
gammascalpel_score_contrasts.csv	TaskGamma and MeaningGamma by band and branch contrast.	Gamma is work/load; artifact veto applies.
temporal_semantic_proxy_window_scores.csv	TSP traces by gamma band and window.	Scalp proxy, not STS/MTG source localization.
postpeak_peak_and_taper_detection.csv	Detected peak and taper anchors by predictor trace and scope.	Physiology-discovered anchors are exploratory.
postpeak_pncc_theta_contrasts.csv	Theta pre/post changes after gamma/TSP taper anchors.	Requires Target specificity and artifact guard.
annotation_locked_theta_carryover_contrasts.csv	Theta carryover after supplied media/scene anchors.	Confirmatory only if predeclared/frozen.
gamma_to_theta_handoff_best_lags.csv	Best lagged gamma/TSP -> theta correlations.	Correlation is not causality.
thresholded_state_update_candidate_windows.csv	State-update candidate grades after taper anchors.	Exploratory unless anchor rule frozen.
human_translation_kht_phasewise_summary.csv	Phase-level eta, zeta, K_HT by branch.	K_HT not K_OSM.
candidate_local_kht_analysis.csv	Event-level local K, theta deltas, percentiles, lock flags.	Lock requires predeclared anchor + local K + theta handoff.
master_endpoint_status_table.csv	One-row-per-endpoint summary.	Always inspect the evidence table before quoting.

10. Worked example: how Her changed the doctrine

- Her Run 1 did not produce a clean phase-level K_HT pass: Target exceeded Control but did not exceed Override at the phase level.
- The originally predicted frozen subjective-time windows were weak.
- An exploratory target-local event around 113.9 seconds showed elevated local K and theta carryover, but because it was physiology-discovered/post hoc in that run it could not confirm itself.
- On later review, the 113-second point mapped to a media-structural event: the letter reading finishes and the characters reach the top of the roof. That makes it an excellent future predeclared anchor for a repeat Her-style test.
- This is the correct lesson: do not ask an absorbed participant to be a stopwatch. Freeze the scene or freeze the algorithm before analysis.

Chronometric humility rule

Freeze the scene, freeze the algorithm, and let the subject describe meaning without pretending to be a clock. Retrospective second-level self-time estimates should not carry confirmatory weight.

11. Allowed and forbidden claims

Allowed	Forbidden
The suite estimates exploratory EEG/autonomic/state-locked endpoints under the PR-AYC-G three-arm design.	The suite proves consciousness, empathy as a force, or OSM.
GammaScalpel decomposes lower-gamma work signals by band, ROI, task, visual, and artifact penalties.	GammaScalpel directly measures meaning.
TSP is a scalp-level posterior-temporal lower-gamma proxy after nuisance penalties.	TSP localizes STS or MTG.
K_HT is a human-translation coupling proxy between E_human and Y_HT.	K_HT is K_OSM or proof of a cellular hidden-Y mechanism.
CandidateLocal_KHT can generate future frozen anchors and can test predeclared anchors.	Physiology-discovered events confirm themselves in the same run.
ALS validates the physical timing input u(t).	ALS measures biological photons or hidden-Y.

12. TSC explanation script

30-second explanation

PR-AYC-G compares a phase-scrambled control, an intact narrative, and the same intact narrative under analytic override. The Master Comprehensive Suite turns the XDF into time-windowed EEG, autonomic, respiration, timing, and self-report features. It then tests whether target narrative processing differs from sensory-only and task-locked alternatives. The newer modules look for state-locked gamma/TSP-to-theta handoffs and exploratory K_HT coupling, but they retain strict boundaries: human K_HT is not OSM proof, TSP is not source localization, and ALS is timing validation only.

90-second explanation

The suite first validates the run: streams, markers, cue schedule, channel map, ALS timing, ratings, and artifact windows. It then computes EEG bandpower and PLV in short windows. GammaScalpel separates lower-gamma into 30-35, 35-40, 40-45, and 30-45 Hz scores, penalizing frontoparietal task lock, visual control, jaw/temporal sentinels, and artifacts. TSP builds a posterior-temporal scalp proxy by subtracting visual/task/jaw/artifact terms. The handoff modules locate a gamma or TSP peak, detect its taper, then ask whether theta increases afterward more in Target than Control or Override. HumanTranslation_KHT estimates a rough reciprocal loop between gamma/work E(t) and human translation proxy Y_HT(t). CandidateLocal_KHT applies the stricter Rung 0O doctrine at event level: K alone is not enough and theta handoff alone is not enough; both must pass, and the anchor must be predeclared or media-structural for a stronger claim.

13. Module glossary

Term	Meaning in suite
E_human(t)	Standardized EEG gamma/work proxy used in K_HT modules.
Y_HT(t)	Human translation proxy, usually TSP or constructed posterior-temporal lower-gamma residual.
eta	Local/phasewise coefficient for Y_HT -> E_human.
zeta	Local/phasewise coefficient for E_human -> Y_HT.
K_HT	sqrt(eta*zeta) for positive reciprocal product; human translation

	proxy only.
TSP	TemporalSemanticProxy; posterior-temporal scalp lower-gamma after nuisance penalties.
PNCC_theta	Post-narrative/coherence/carryover theta family.
PostPeak PNCC	Theta carryover after a detected gamma/TSP peak taper, rather than after video end.
State-locked	Aligned to a physiological or media-event boundary rather than the MP4 start/end alone.
Media-structural anchor	Predeclared scene/time event in the rendered stimulus, independent of physiology.
Artifact score	Average z-scored p2p/high-frequency sentinel features used to penalize noisy windows.

14. Closing boundary

Final technical boundary

The suite is designed to sharpen the scalpel. It is not designed to let one run, one metric, or one striking figure outrun the evidence. A clean PR-AYC-G result must survive branch comparison, artifact veto, timing validation, self-report coherence, and simpler-model alternatives.

Appendix A. Source and lineage notes

- This handout describes PRAYCG Master Comprehensive Suite v1.4.0 and MasterSync Visualizer v1.2.0 as packaged in the unified v1.4 operational bundle.
- The PR-AYC-G branch rule is carried forward from the MediaPrep lineage: Target and Contextual Override use the same cue-embedded stimulus; Control is phase-scrambled from the cue-embedded Target.
- The ALS timing doctrine is carried forward from the ALS-PT19 QC lineage: ALS validates physical display timing and belongs to u(t), not biological Y(t).
- The K_HT doctrine is carried forward from the Opto-PING/Rung 0O lineage: prediction alone is not enough, K alone is not enough, and human K_HT is not K_OSM.
- This document intentionally distinguishes engineering outputs, exploratory evidence, confirmatory anchors, and mechanism claims.

Appendix A - Custom Composite Metrics and Decision Rules (v1.2 Formula Addendum)

Purpose. This appendix exposes the custom PR-AYC-G formulas that are not standard signal-processing boilerplate. Welch PSD, Hilbert transforms, and ordinary PLV are deliberately kept in the methods prose; the formulas below document the suite-specific composites that reviewers need in order to audit the logic.

Implementation status. These are the v1.4/v1.2 documentation defaults. If future code changes a weight, threshold, reference window, or veto rule, this appendix should be updated with the code in the same versioned release.

Boundary. These formulas produce exploratory psychophysiological indices. They do not certify meaning, consciousness, OSM, hidden-Y biology, or a cellular mechanism. They make the analysis more inspectable; they do not make the interpretation automatic.

A1. Robust positive z-score used by custom composites

Many suite metrics use robust reference scaling so that one noisy block does not dominate the feature space. The positive-only transform is used for artifact features because unusually clean windows should not produce a negative artifact bonus; they should simply carry little or no penalty.

```
z_plus(x_w) = max(0, (x_w - median(x_ref)) / (1.4826 * MAD(x_ref) + epsilon))

where:
  x_w     = current window value
  x_ref   = reference distribution, usually baseline or valid non-report windows
  MAD     = median absolute deviation
  epsilon = small numerical stabilizer
```

- Use positive-only z-scoring for artifact burden, not for all scientific endpoints.
- A low artifact score means the feature is less penalized; it does not mean the window is meaningful.

A2. artifact_score(w)

The artifact score compresses sentinel-channel amplitude excursions and high-frequency contamination into one penalty term. It is meant to reduce the influence of jaw tension, movement, electrode disturbance, and high-frequency muscle-like contamination in gamma-sensitive modules.

```
P2P_s(w) = Q95(x_s[w]) - Q5(x_s[w])
HF_s(w)  = log(Power_45_55_s(w) + epsilon)

ArtifactScore(w) = [ alpha_p2p * mean_s z_plus(P2P_s(w))
                   + alpha_hf  * mean_s z_plus(HF_s(w)) ]
                   / (alpha_p2p + alpha_hf)

Default v1.2 documentation weights:
  alpha_p2p = 0.5
  alpha_hf  = 0.5
```

The default feature penalty is written as:

```
Penalty_A(w) = 1 / (1 + lambda_A * ArtifactScore(w))

CleanFeature(w) = RawFeature(w) * Penalty_A(w)
```

- P2P captures large voltage excursions and contact/movement instability.
- HF_45_55 captures high-frequency contamination near the lower-gamma/high-beta border where muscle and line-like noise can become dangerous.
- The score is a penalty/veto layer, not a meaning detector.

- If artifact_score is high at the same time as gamma/TSP, the interpretation is downgraded or rejected unless additional controls explain why the event is still credible.

A3. API_A_v1(w): autonomic availability proxy

API_A_v1 is a custom autonomic state proxy designed to distinguish regulated availability from mere arousal. It combines heart-rate slowing/stability and variability features into one directionally interpretable index. It is not a clinical diagnostic index and should not be interpreted without respiration, task, and artifact context.

```
API_A(w) = [ sum_{q in Q_plus} a_q * z(q_w)
             - sum_{r in Q_minus} b_r * z(r_w) ]
             / [ sum_{q in Q_plus} a_q + sum_{r in Q_minus} b_r ]

Default feature sets:
  Q_plus  = [RMSSD, SDNN]
  Q_minus = [HR, abs(dHR/dt)]
```

Equal-weight default:

```
API_A_v1(w) = [ z(RMSSD_w) + z(SDNN_w) - z(HR_w) - z(abs(dHR/dt)_w) ] / 4

If SDNN is unavailable in a short window:
API_A_v1(w) = [ z(RMSSD_w) - z(HR_w) - z(abs(dHR/dt)_w) ] / 3
```

- Higher API_A_v1 means relatively lower HR strain, higher HRV, and less abrupt HR acceleration within the chosen reference scaling.
- Lower API_A_v1 means relatively higher autonomic load or less regulated availability.
- API_A_v1 is interpreted alongside respiration because respiration can drive HRV and can mimic or obscure autonomic-neural coupling.
- API_A_v1 should not be treated as an emotion score, empathy score, or parasympathetic truth score.

A4. TemporalSemanticProxy: construction, peak detection, and half-taper rule

The TemporalSemanticProxy (TSP) is a scalp-level posterior-temporal semantic-work proxy. It is not STS/MTG source localization. It asks whether posterior-temporal lower-gamma activity rises beyond visual-control and artifact burden in a way that could mark semantic/narrative processing.

```
TSP(t) = w_T * mean_{ch in TemporalROI} z(Gamma_30_45_ch(t))
       - w_V * mean_{ch in VisualControlROI} z(Gamma_30_45_ch(t))
       - w_A * ArtifactScore(t)

Default interpretation:
  TemporalROI      = posterior-temporal scalp proxy channels, if channel map is locked
  VisualControlROI = occipital/visual-control channels
  Gamma_30_45      = lower-gamma band used as semantic-work candidate
  ArtifactScore    = artifact penalty from A2
```

- The visual-control subtraction is intended to reduce the chance that the proxy is simply luminance, motion, or raw visual drive.
- The artifact penalty is intended to reduce the chance that the proxy is jaw/temporal muscle, movement, or electrode disturbance.
- TSP is strongest when temporal gamma rises while visual-control and artifact burden do not explain the event.

A4.1 TSP peak detection

```
t_peak = argmax_t TSP(t)
```

```
Candidate peak is accepted only if:
  TSP(t_peak) > Q_TSP_threshold
  ArtifactScore(t_peak) < A_max
  t_peak is not inside a censored report/input period
  t_peak obeys the refractory rule R seconds from prior accepted peaks
  branch/condition filter is satisfied
```

- Q_TSP_threshold can be a robust percentile threshold or a preregistered z threshold depending on run mode.
- The refractory rule prevents one extended bump from becoming many separate events.
- If the anchor is physiology-discovered, it remains exploratory unless the detection rule was frozen before analysis.

A4.2 Half-taper threshold and taper start

The handoff logic does not ask only whether TSP peaks. It asks whether theta rises after the semantic-work signal has tapered, because the model treats theta as an integration/carryover candidate rather than the same event as the gamma/TSP intake peak.

```
TSP_base = median(TSP(t_ref))

TaperThreshold = TSP_base + rho * [TSP(t_peak) - TSP_base]

Default:
  rho = 0.5

TaperStart = min { t > t_peak : TSP(t) < TaperThreshold for at least d_min seconds }
```

- If TSP never falls below the taper threshold, the module reports no valid taper rather than forcing a handoff window.
- A taper is a timing anchor, not proof of meaning.

A4.3 TSP-to-theta handoff

```
H_TSP_to_theta = mean(theta_z[TaperStart + tau_1 : TaperStart + tau_2])
               - mean(theta_z[t_peak - b : t_peak])

Typical exploratory defaults:
  pre window b          = 10 to 15 seconds
  theta carryover tau_1 = 0 to 10 seconds after taper, or 10 seconds after peak
  theta carryover tau_2 = 30 seconds after taper/peak
```

- A positive handoff means theta increased after the TSP peak/taper relative to the pre-event window.
- The result is tested against random-anchor or matched-window nulls, not interpreted from a raw positive number alone.

A5. CandidateLocal_KHT event model

CandidateLocal_KHT is an event-level diagnostic. It asks whether, around a candidate anchor, a gamma-like work signal and a TSP-like human-translation signal show local reciprocal coupling. It is a human-translation proxy and must not be renamed K_OSM.

```
For candidate event j:

E_t        = gamma_z(t)
Y_HT,t     = TSP_z(t)
Artifact_t = ArtifactScore(t)
```

```
Local E equation:
E_{t+1} = a_E * E_t + eta_j * Y_HT,t + c_E * Artifact_t + epsilon_E,t

Local Y_HT equation:
Y_HT,{t+1} = a_Y * Y_HT,t + zeta_j * E_t + c_Y * Artifact_t + epsilon_Y,t

K_local,j = sqrt(abs(eta_j * zeta_j))
s_j       = sign(eta_j * zeta_j)
```

- eta_j is the local Y_HT -> E coefficient inside the candidate window.
- zeta_j is the local E -> Y_HT coefficient inside the candidate window.
- K_local,j is the magnitude of the reciprocal event loop. The sign s_j is tracked separately because reinforcing and opposing loops are not equivalent.
- This is an exploratory local regression model, not a cellular mechanism measurement.

A6. CandidateLocal_KHT theta handoff and final human-event lock rule

The suite uses the same conservative discipline learned from the synthetic Rung 0O gate: prediction alone is not enough, and K alone is not enough. A human event requires local K evidence, theta carryover, specificity, artifact control, and timing validity.

```
Theta handoff for candidate j:

H_theta,j = mean(theta_z[t_j + tau_1 : t_j + tau_2])
          - mean(theta_z[t_j - b : t_j])

Human event-lock rule:

EventLock_j = 1[K_local,j > Q_K,null]
            * 1[H_theta,j > Q_theta,null]
            * 1[Artifact_j < A_max]
            * 1[Target_j > Control_j]
            * 1[Target_j > Override_j]
            * 1[TimingStatus_j passes]
```

Plain-language rule:

```
K alone is not enough.
Theta handoff alone is not enough.
Target-only activity is not enough if artifact, timing, or respiration explains it.
Final human-event lock requires local coupling + theta carryover + condition specificity +
artifact/timing pass.
```

- If the event is discovered after looking at physiology, it is marked physiology_exploratory and cannot be confirmatory in the same run.
- If the event is a predeclared media-structural anchor loaded before acquisition or before physiological analysis, it can be tested as a confirmatory anchor, provided timing and QC pass.
- If self-report identifies a scene after the run, it is a scene-level qualitative constraint, not a precise clock source by itself.

A7. How these formulas enter the module reports

The v1.2 formulas map directly onto output tables and interpretation fields:

Formula block	Primary outputs	Interpretation role

artifact_score	artifact_score columns, artifact veto fields, cleaned feature columns	Penalizes gamma/TSP/KHT windows likely driven by movement, jaw, contact, or high-frequency contamination.
API_A_v1	api_a, hr, rmssd, sdnn, hr_slope fields	Summarizes regulated autonomic availability; interpreted with respiration/task context.
TSP peak/taper	tsp_peak, taper_start, tsp_to_theta_handoff, state-locked anchor tables	Locates candidate semantic-work peaks and tests whether theta follows the taper.
CandidateLocal_KHT	candidate_local_kht_analysis.csv, visual overlay CSV, interpretation JSON	Tests local gamma/TSP reciprocal coupling and event-lock criteria.

A8. Version 1.2 claim boundary

These equations make the Master Comprehensive Suite easier to audit. They do not change the empirical hierarchy of claims:

- PR-AYC-G remains a psychophysiology protocol about whether meaning changes measurable state beyond light, sound, pacing, task effort, and noise.
- ALS/PT19 timing validates the physical stimulus input u(t); it does not measure biological Y(t).
- CandidateLocal_KHT is a human-translation event diagnostic; it is not K_OSM.
- Rung 0 synthetic results validate a simulation/reproducibility pipeline; they do not prove human EEG mechanism or OSM biology.
- The strongest scientific result is one that can say no when artifact, timing, respiration, or controls explain the pattern.